

Dustin Nguyen

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Summary

AI/ML Software Engineer with Physics PhD and 2+ years building and deploying production ML systems across defense applications. Currently working in the Autonomous Systems team at  Palantir.

Technical Skills

Tools: Python, PyTorch, git, Lightning, Weights & Biases, Mlflow

Cloud & Deployment AWS EC2 and S3, Data pipelines, Kafka, Docker, TensorRT, Model Quantization

ML Research: Continuous-time SciML models, Transformers, Agentic Workflows

Non-ML Domains: Flight Dynamics, Space Domain Awareness, Physics, Numerical Methods

Experience

Software Engineer Palantir Technologies – Seattle, WA

Apr 2026 – Present

- *applied research in the autonomous systems team. more to come later*

Senior Machine Learning Research Scientist, Leidos – Remote, USA

Mar 2025 – Apr 2026

- Architected Transformer time-series surrogate model for DARPA enabling real-time Space Domain Awareness at constellation scale - **1000x faster and 30% relative reduction in error** than legacy physics-based methods.
- Designed **sim-to-real training** pipeline and novel feature engineering to ensure generalization under limited real-world data; owned end-to-end ML workflow using Mlflow.
- Productionized ML model by containerizing with Docker and integrating into a Kafka streaming pipeline, implementing producer/consumer workflows for real-time inference at scale.
- Contributed to an IARPA remote-sensing program, an Edge-AI IRAD, and solutioning towards capture efforts.

Senior Machine Learning Engineer, Lockheed Martin – Denver, CO

Mar 2024 – Mar 2025

- Led Python model development for a DARPA project deploying Controlled Neural ODEs as surrogate models in AFSIM simulations. Scaled distributed memory-efficient adjoint-based training with PyTorch Lightning across multi-GPU systems and integrated models into AFSIM using LibTorch. Delivered quantitative results and analyses to government stakeholders through monthly SharePoint presentations.
- Contributed to an IRAD focused on multi-modal classification.

Post-doctoral Researcher, Ohio State University – Columbus, OH

Dec 2023 – Mar 2024

Applied Machine Learning Fellow, Los Alamos National Laboratory – Los Alamos, NM

May 2022 – Aug 2022

- Researched and developed Neural ODEs as a surrogate model for non-linear physics using Julia language.

Graduate Researcher & NASA FINESST Fellow, Ohio State University – Columbus, OH

Aug 2020 – Dec 2023

- Studied idealized galactic feedback models, which is relevant to understanding galaxy evolution, using theoretical and computational methods. Total of 10 published papers, with 6 being first author.
- Awarded competitive NASA FINESST Fellowship (acceptance rate ~8%) for proposed research

Projects

Universal Differential Equations for learning non-linear physics

- Developed the first use case of Neural ODEs (UDEs) within idealized galactic feedback models using Julia.
- Published proof-of-concept model into ICML 2023 Workshop on ML for Astrophysics.
- Applied to real-world data and published into NeurIPS 2023 Workshop on ML and the Physical Sciences.

Courses

DeepLearning.ai

- Finetuning & Reinforcement Learning (RL) for LLMs: SFT, RLHF, GRPO, Constitutional AI, etc.
- Retrieval Augmented Generation (RAG): Retriever, Vector Databases, Rerankers, Chunking, etc. (2025 [↗](#))
- Agentic AI: Multi-agent Workflows, Task Decomposition, Tools, Reflection, Evaluations, etc. (2025 [↗](#))
- Machine Learning Specialization: ML Fundamentals (2023 [↗](#))

Education

Ohio State University – PhD in Physics	Dec 2023
Ohio State University – MS in Physics	May 2021
Arizona State University – BSs in Physics and Astrophysics	May 2018